



School of Computer Science

A Level 1 Module COMP1038 Autumn 2024/25

Programming and Algorithms

Instructions:

Time allowed: One Hour

Answer ALL questions

Candidates may complete the front cover of their answer book and sign their desk card. **DO NOT turn the examination paper over until instructed to do so.**

Permitted resources:

Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination.

No calculators are permitted in this examination.

Prohibited resources:

All other dictionaries, including subject specific translation dictionaries and electronic dictionaries. Electronic devices capable of storing and retrieving text.

Exam Papers must not be removed from the examination venue.

Question 1: What is the reason and consequence of memory leaks in C?

[2 Marks]

Question 2: What is a pointer to a pointer in C? Write a C program that uses a pointer to a pointer to change the value of an integer variable.

[5 Marks]

Question 3: Error Identification and Correction

The following C program is intended to read a list of integers from the user, calculate their sum, and print the result. However, it contains several errors. Identify and correct these errors. Provide the corrected code and explain each correction you made.

[4 Marks]

```
1  #include <stdio.h>
2  void main() {
3      int n, i, sum;
4      printf("Enter the number of elements: ");
5      scanf("%d", n);
6      int arr[n];
7      printf("Enter %d integers:\n", n);
8      for(i = 0; i <= n; i++) {
9          scanf("%d", &arr);
10     }
11     sum = 0;
12     for(i = 0; i < n; i++) {
13         sum = arr[i];
14     }
15     printf("Sum of the elements: %d\n", sum);
16 }
```

Question 4: Array Manipulation and Sorting

[6 Marks]

Problem:

Write a C program that reads 10 integers into an array, sorts the array in ascending order, and then print the sorted array to the screen.

Instructions:

- Read 10 integers from the user into an array.
- Sort the array in ascending order using any sorting algorithm.
- Print every item in the sorted array.

Question 5: Write a C program that counts the number of English alphabetic characters in a given string. The program should prompt the user to input a string, then iterate through each character to count how many are English letters (both uppercase and lowercase). Finally, it should display the total count of English alphabetic characters. The program must not use any built-in string manipulation and character handling functions. You can assume that the input string will contain no more than 1000 characters.

[8 Marks]

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